Single neuron and population analysis of visual motion direction : effect of uncertaincy.

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Introduction

Studying the **internal representation** of natural features in the primary visual cortex (V1), like direction or spatial frequency, is crucial to understand how we perceive the external world. Research on 2D motion direction in **non-human primates** in particular when displaying **naturalist-like stimuli** like MotionClouds reveals substantial diversity and multiple mechanisms within the neuronal population [1]. The aim of this project is to examine how a large population of V1 neurons encodes stimulus direction and how this representation is modulated by the uncertainty using classical and decoding methods.

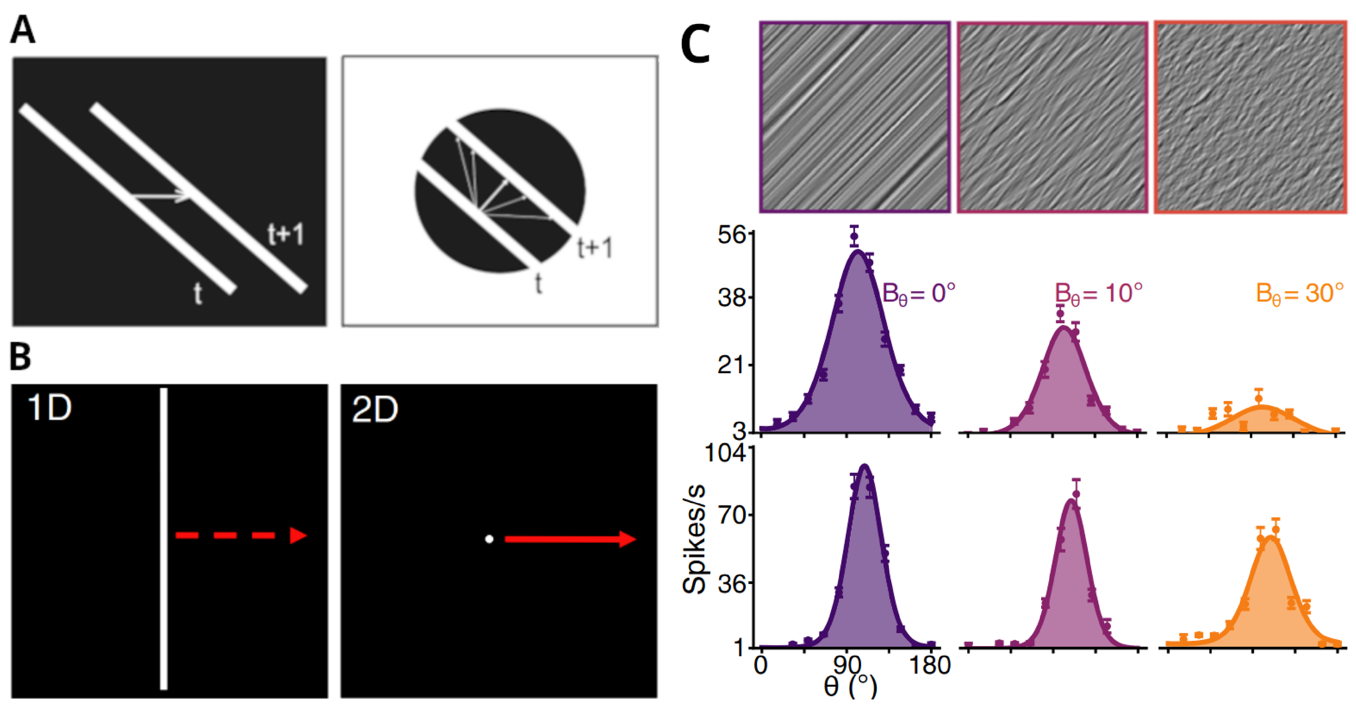

Figure 1: **A.** The aperture problem : information about motion is ambiguous when it is observed through a small aperture. **B.** Moving stimuli carrying mostly 2D (central spot, right) or 1D (very long line, left) information. **C.** From 1D to 2D, two subpopulation in cat area 17. (A and B adapted from [2]; C adapted from [1])

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Methods

Neuronal recordings were made in V1 of one marmoset monkey using the Neuropixel technology during 2D motion presentation in twelve directions (θ) from $0°$ to $375°$, four levels of orientation bandwidth (B_θ; $0°$, $15°$, $30°$ and $45°$) and 4 levels of spatial frequency bandwidth (B_{sf}; 0.01, 0.175, 0.7 and 2.8 cpd) from the MotionClouds python library.

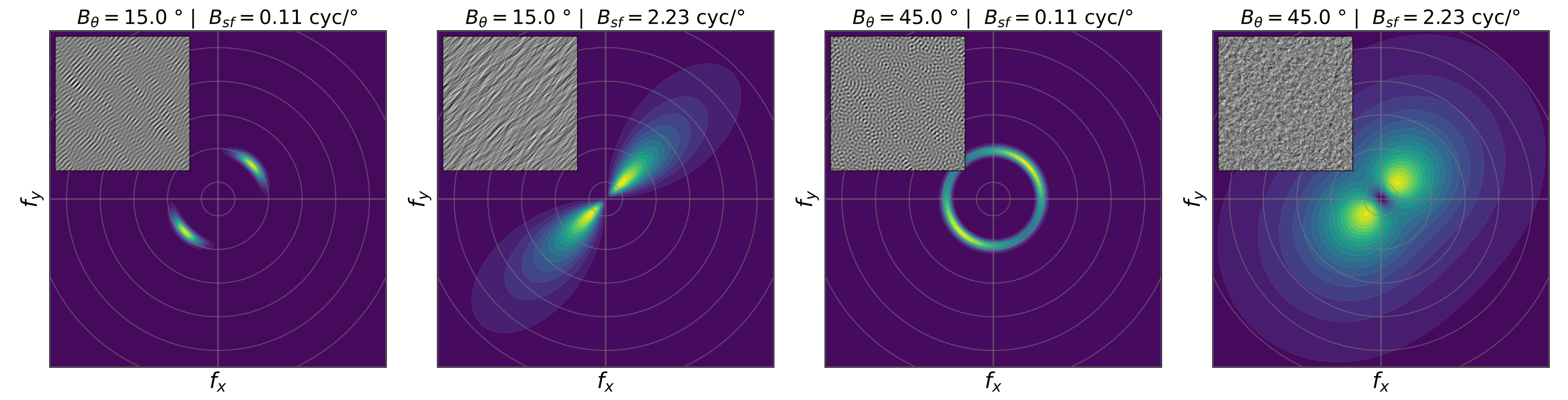

Figure 2: Fourier space representation of the envelope of MotionClouds stimuli. Resulting images (top-left inset) are generated by the inverse Fourier transform of the envelope after the application of a random phase.

The 2D (f_r; f_θ) functional space used is described as follows The idea is to find the best template in Fourier space that predicts the neuronal activity induced by each stimuli.

$$f_r = e^{-\frac{1}{2}\log\frac{sf - sf_0}{B_{sf}}}$$
$$f_\theta = e^{\frac{\cos 2(\theta-\theta_0)}{4B_\theta^2}}$$

The tuning curve model is defined by the following equations in order to predict the activity (A) of each neuron in each condition.

$$A_\theta = \kappa_\theta \times (\cos(2\times(\theta-\theta_0)) - 1)$$
$$A_\phi = \kappa_\phi \times (\cos(\theta-\theta_0) - 1)$$
$$A = e^{A_\theta + A_\phi + \log R_{amp}} + R_0$$

Orientation and direction selective indices (OSI and DSI) are computed from our parameters (κ_ϕ, κ_θ, θ_0...).

$$DSI = \frac{R_{\text{pref}} - R_{\text{null}}}{R_{\text{pref}}} = \frac{R_{\text{amp}}(1 - e^{-2\kappa_\phi})}{R_0 + R_{\text{amp}}}$$

Decoding method based on [3] define the likelihood function of the stimulus direction as an equivalent to the sum of the likelihood functions of each neuron, which is equivalent to summing the agreement functions of each neuron weighted by the activity of each neuron during stimulus presentation. If we take W_i as the tuning curve of neuron i and A_i^s as the activity induced by simulus s. Then, L_i^s and L^s are respectively the likelihood function for a particular trial of the neuron i and the population.

$$\log L^s(\theta) = \sum_{i=0}^{N} \log L_i^s(\theta) = \sum_{i=0}^{N} A_i^s(\theta)\log W_i(\theta)$$

Results : single neuron analysis

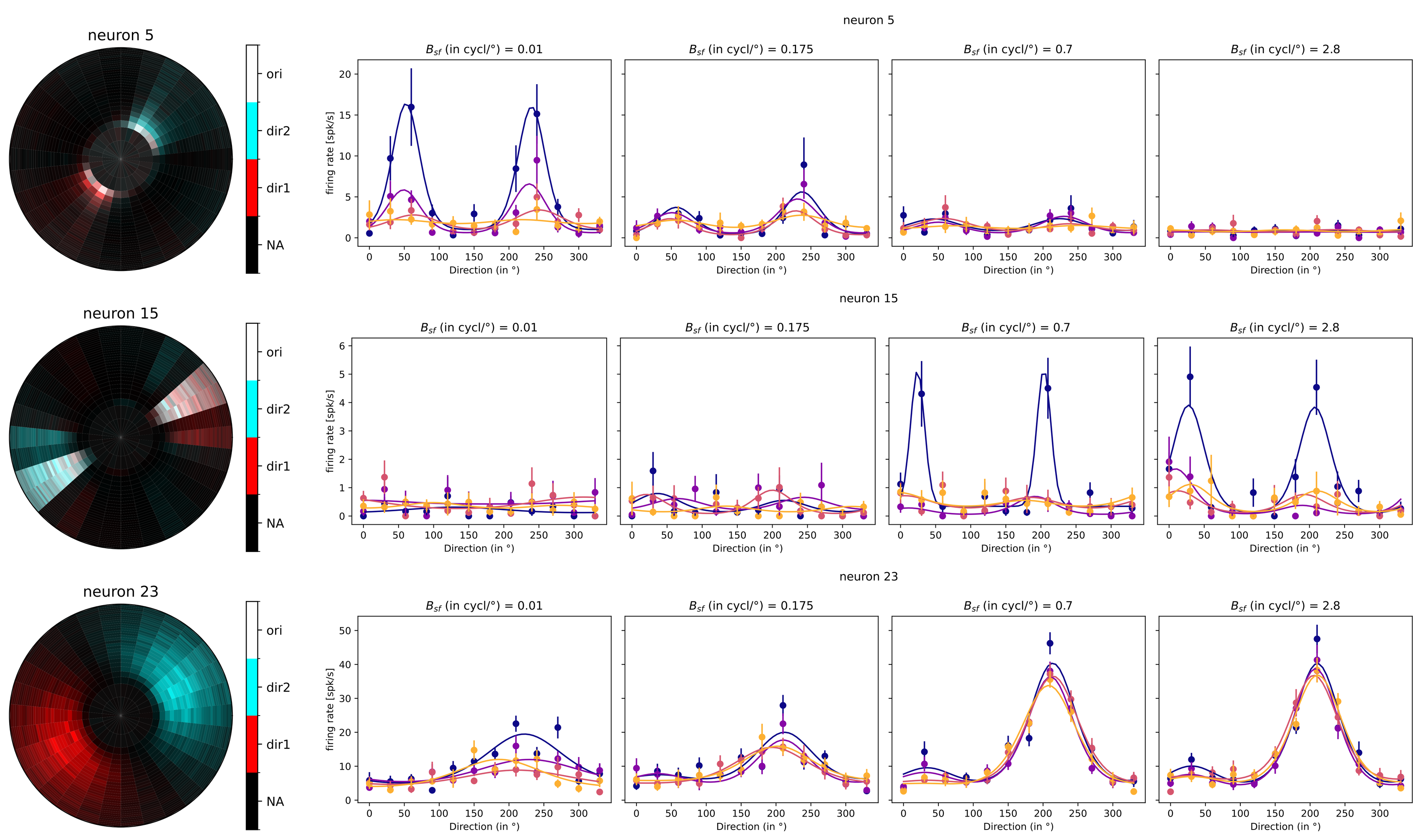

Figure 3: Mean DSI and OSI in population.

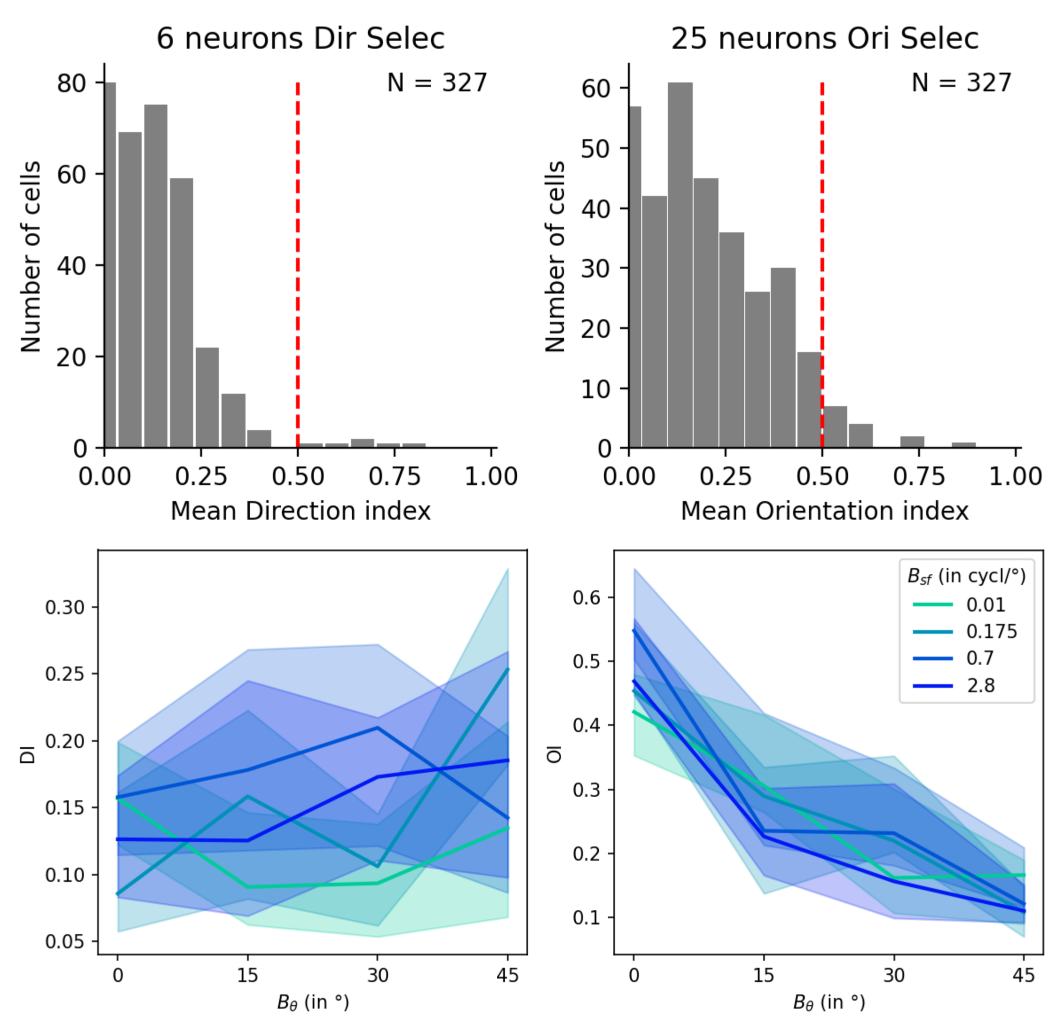

Figure 4: Mean DSI and OSI in population.

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Results : population decoding

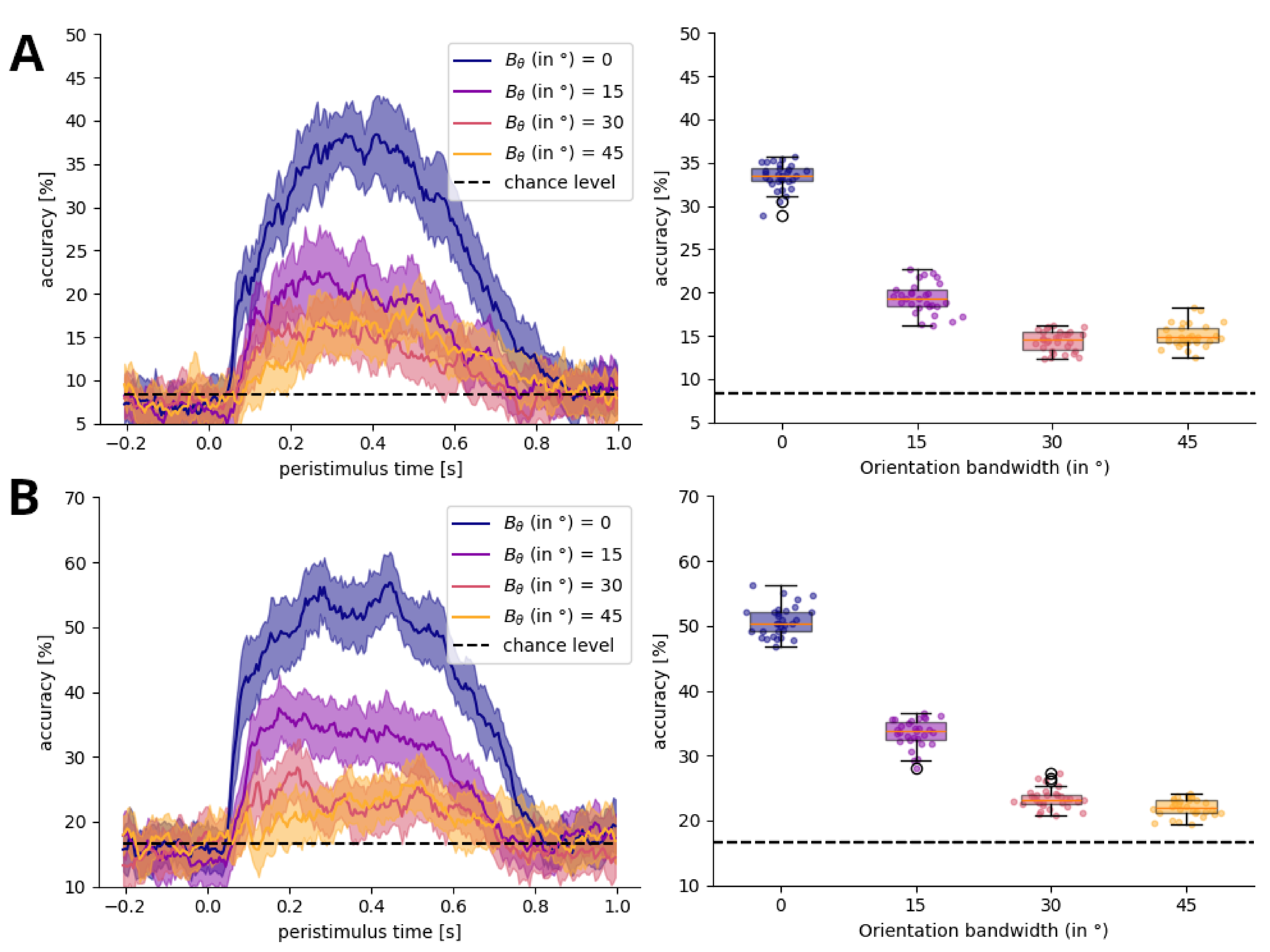

Figure 5: Decoding results for multiple level of orientation bandwidth. **A.** Decoding the direction of the stimulus. **B.** Decoding the orientaiton of the stimulus.

Discussion

The encoding approach This decoding approach clarifies the **representation of directional information** in the marmoset primary visual cortex.

References

[1] Hugo J. Ladret et al. “Cortical recurrence supports resilience to sensory variance in the primary visual cortex”. In: *Communications Biology* 6.1 (2023), p. 667.
[2] Anna Montagnini et al. “Bayesian modeling of dynamic motion integration”. In: *Journal of Physiology-Paris* 101.1–3 (Jan. 2007), pp. 64–77.
[3] Mehrdad Jazayeri and J Anthony Movshon. “Optimal representation of sensory information by neural populations”. In: *Nature Neuroscience* 9.5 (May 2006), pp. 690–696.